

Operating Systems

Tutorial Worksheet 04

Description. This tutorial worksheet covers the topic of process synchronization.

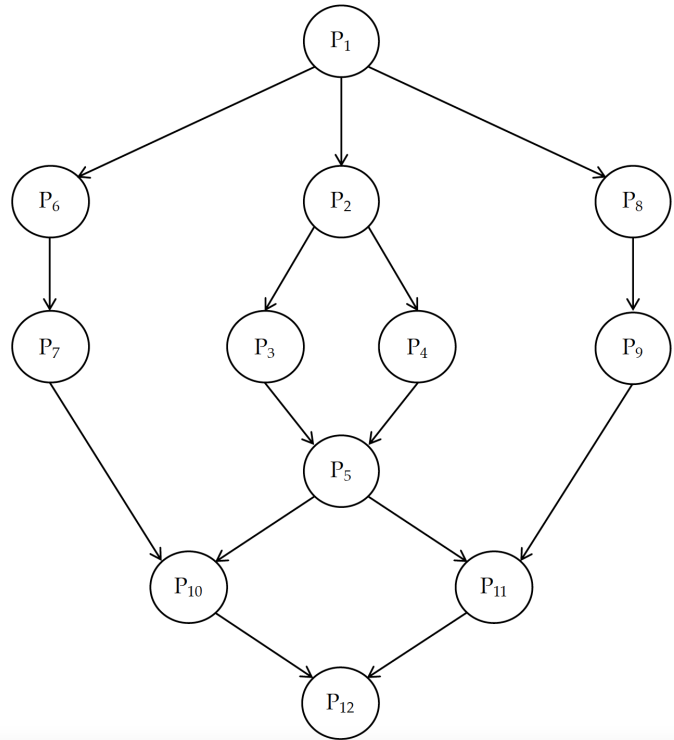
1. Process Precedence Graph

- a) Draw a process precedence graph (a.k.a., dependency graph) that reflects the parallelism in the following code (viz., figure below-left). Recall a precedence graph is a directed graph in which the nodes (vertices) represent processes and the edges represent the causal relation between two processes.

```

CoBegin
  P0 ;
  Begin
    P1 ;
    P2 ;
  End
  P3 ;
CoEnd
P4 ;
P5 ;
CoBegin
  P6 ;
  Begin
    P7 ;
    CoBegin
      P8 ;
      P9 ;
    CoEnd
    P10 ;
  End
CoEnd
P11 ;

```



- b) Write a code using the CoBegin and CoEnd construct to execute the processes illustrated in the given PPG (viz., figure above-right), while respecting all the precedence relations. You may consider introducing the least possible additional dependencies and the least extra delays. If extra dependencies do occur, explain where and why.
- c) For the two PPGs in parts (a) and (b), assume that all processes execute concurrently. Implement the required precedence relations using the minimum possible number of semaphores, assuming no other variables nor structures are used. To show your answer, you must first plot your assignment of semaphores to edge on the graph and then write the code for each process using only P(), V(), and critical sections (CS).

- d) (Optional) In the question of part (b), if each process P_i needs i units of time from start to finish of its execution, can you utilize this fact rewrite the code in part B to makes sure that no unnecessarily precedence relations or excessive delays are endured by any of the processes (Hint: You may relax some of the actual precedence relations in PPGs as long as they are respected time wise).

2. Process Synchronization using Semaphores

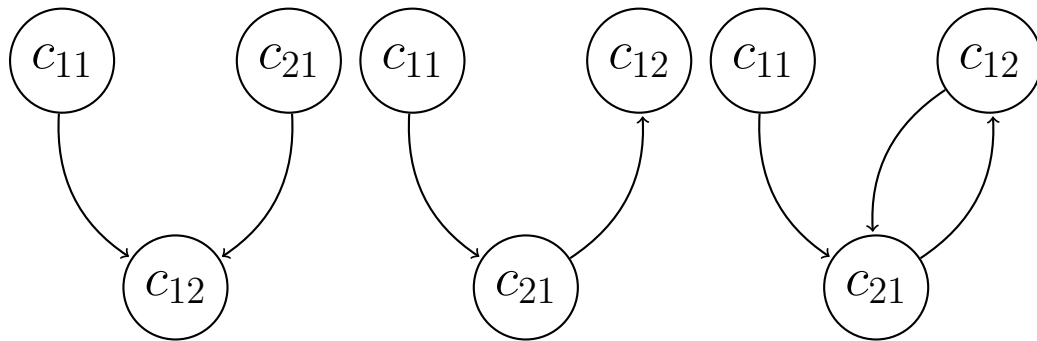
The shown codes are for two processes that are executing concurrently in a system. Process P_1 has two critical sections, whereas P_2 has only one. Complete the codes of the processes using the minimum number of semaphores and any other basic programming approaches so as to respect the following precedence relations of their critical sections.

P1:

```
Begin
  CS11;
  CS12;
End
```

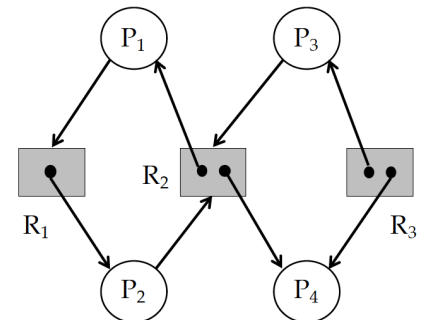
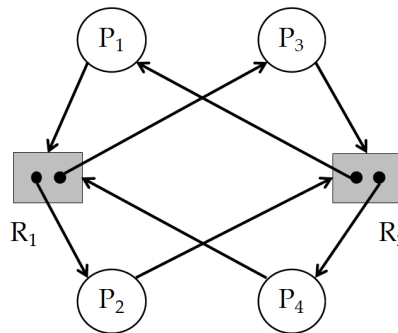
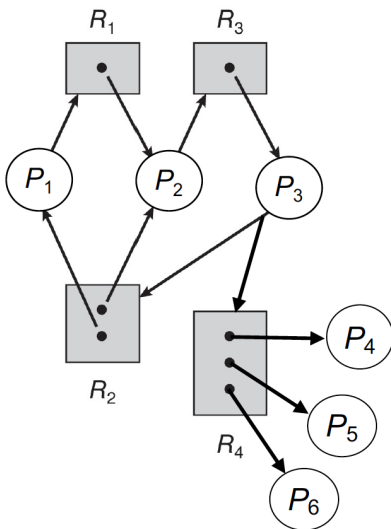
P2:

```
Begin
  CS21;
  nop;
End
```

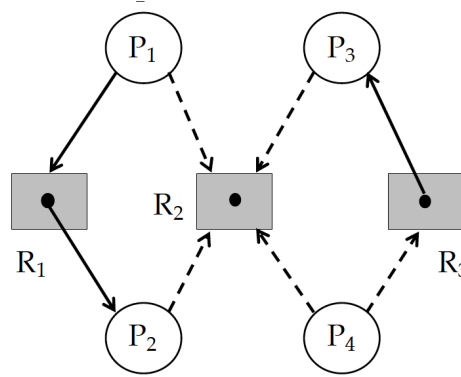


3. Deadlock Handling

- a) Assume that the mutual exclusion, hold-and-wait, and no-preemption conditions are all true for the systems represented by the shown RAGs (Resource Allocation Graphs). Identify whether each of them is in a deadlocked state or not. Justify your answer with a clear description of the situation.



- b) Again assume that the mutual exclusion, hold-and-wait, and no-preemption conditions are all true for the systems represented by the shown RAG, identify whether a system implementing the RAG algorithm will grant any of the following requests. Justify your answer with schematics and description of the situation.



- i. P_1 requests R_2 .
 - ii. P_2 requests R_2 .
 - iii. P_4 requests R_2 .
 - iv. P_4 requests R_3 after P_3 releases R_3 and acquires R_2 .
- c) The table below depicts the allocated instances of 4 resource-types A,B, C, and D by 5 processes at a given time t_1 . The table also depicts the maximum no. of resources each of these processes may need in the near future to complete the execution its code. Assuming that the system has 6, 28, 24, and 24 instances of resource-types A, B, C, and D, respectively.

<u>Processes</u>	<u>Allocation</u>				<u>Max</u>			
	A	B	C	D	A	B	C	D
P_1	2	6	10	8	4	6	10	12
P_2	0	0	3	8	0	17	10	12
P_3	2	0	0	0	2	18	22	0
P_4	0	0	1	4	0	0	2	4
P_5	0	12	6	4	0	12	14	4

- i. Demonstrate that this system is in a safe state.
- ii. If at time t_1 , P_3 makes a requests to the operating system to acquire 8 and 4 instances of resource types B and C, respectively, will the operating system grant this request immediately? Justify your answer with full details.
- iii. If at time t_1 , P_2 makes a requests to the OS to acquire 10, 4, and 4 instances of resource types B, C, and D, respectively, will the OS grant this request immediately? Justify your answer with full details.
- iv. If at time t_1 , P_5 makes a requests to the OS to acquire 1, and 4 instances of resource types A and C, respectively, will the OS grant this request immediately? Justify your answer with full details.